Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/16/2012
PI Name:	Mulavara, Ajitkumar P. Ph.D.		
Project Title:	Development of Countermeasures to Aid Functional Egress from the Crew Exploration Vehicle Following Long Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Formerly at Universities Space Research Association		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	10/01/2009	**End Date:**	05/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: New end date is 05/31/2014 per NSBRI (Ed., 10/22/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bloomberg, Jacob (NASA Johnson Space Center) Cohen, Helen (Baylor College of Medicine) Reschke, Millard (NASA Johnson Space Center) Wood, Scott (NASA Johnson Space Center) Peters, Brian (Wyle Laboratories)		
Grant/Contract No.:	NCC 9-58-SA02001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Crewmember adapted to the microgravity state may need to egress the vehicle within a few minutes for safety and operational reasons after g-transitions. During exploration class missions the interactions between a debilitated crewmember during re-adaptation to gravity and the prevailing environmental constraints imposed during gravitational transitions may lead to disruption in the ability to perform functional egress tasks. At present, no operational countermeasure has been implemented to mitigate this risk. Therefore, the overall goals of this project are to: 1) investigate performance of motor and visual tasks during simulated perturbation conditions and 2) to develop a countermeasure based on stochastic resonance to enhance sensorimotor capabilities with the aim of facilitating rapid adaptation during gravitational transitions following long-duration spaceflight. Stochastic resonance (SR) is a mechanism whereby noise can assist and hence enhance the response of neural systems by detecting sub-threshold signals. SR thus enables the enhanced detection of relevant sensory signals. SR stimulation using imperceptible noisy vibratory or electrical stimulation has been shown to improve balance function in normal young and elderly subjects, stroke patients and in the rehabilitation of functional ankle joint instabilities. This project specifically has used imperceptible levels of electrical stimulation of the vestibular system (Vestibular SR/VSR) as the proposed countermeasure to improve performance in egress tasks. The project has also conducted a series of studies to document human visual performance during simulated low frequency dynamic perturbations and further investigate the efficacy of VSR stimulation on physiological and perceptual responses during otolith-canal conflicts and dynamic perturbations. The objective of two separate studies that were conducted was to document human visual performance during simulated low frequency vehicular motions. In the first study we examined the changes in accuracy in normals when performing a seated visual target acquisition task in which the location of target was offset vertically during horizontal full-body rotation at an oscillating frequency of 0.8 Hz (peak velocity-160 deg/s). The main finding was that the accuracy of performance degraded by one step size and response times varied with target location when acquiring off-plane targets at perturbing frequencies of 0.8 Hz in the horizontal plane. In the second study we examined 112 normals and 45 patients who had been diagnosed with either unilateral vestibular weaknesses or with post-acoustic neuroma resections. We determined their seated dynamic visual acuity (DVA) task performance while the chair was stationary and during vertical full-body oscillations at perturbing frequencies of 2 Hz (peak-to-peak motion of 5 cm). Scores were worse for both groups during the dynamic condition compared to the static condition. Further, in the dynamic condition patients' scores were significantly worse than normals' scores. These results indicate that low frequency perturbations in the horizontal and vertical planes can cause decrements in visual performance and can exacerbate sensorimotor deficits after spaceflight at low frequencies of dynamic movements. We studied the efficacy of low imperceptible levels of stochastic electrical stimulation of the vestibular system in three separate studies * In the first study VSR efficacy was evaluated during an otolith-canal conflict scenario in a variable radius centrifuge at low frequency of oscillation (0.1 Hz) on physiological responses (using eye movements) and perceptual responses (using a joystick) to track imposed oscillations. Results show that VSR stimulation significantly reduced the phase difference between both the eye counter roll movements as well as the perceptual tracking responses with respect to the imposed tilt in gravito-inertial vector. These results indicate that stochastic electrical stimulation of the vestibular system can improve otolith specific responses in intra-vestibular conflict scenarios like that experienced during spaceflight. * In the second study VSR efficacy was evaluated on the cross-planar improvement in balance performance on an unstable surface while stimulating in the mediolateral (ML) axis only. Low imperceptible levels of white noise based binaural bipolar electrical stimulation of the vestibular system improved cross-planar balance performance. The amplitude of optimal stimulus for improving balance performance was predominantly in the range of 30-210 μA RMS. These results indicate that bipolar binaural stimulation may be sufficient to provide a comprehensive countermeasure approach for improving postural stability. * In the third study VSR efficacy was evaluated during locomotion on an unstable surface. Low imperceptible levels of white noise based binaural bipolar electrical stimulation of the vestibular system improved locomotor performance consistent with the SR phenomenon in normal healthy subjects. These results indicate that VSR may be used to maximize locomotor performance during dynamic movements. The data obtained in this project will aid in the design of a countermeasure system used for improving functional tasks during and after g-transitions. We have also been working with other PIs in the HRP and NSBRI Sensorimotor team. The VSR methodology developed in the current project is being integrated with the sensorimotor adaptability (SA) training modalities being developed by Dr. Bloomberg and his team to improve its efficacy. The operational version of this countermeasure will be available as a skin patch vestibular prosthesis during spaceflight that will further act synergistically along with the pre-and in-flight SA training and provide an integrated, multi-disciplinary countermeasure capable of fulfilling multiple requirements making it a comprehensive and cost effective countermeasure approach.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research Impact: We are proposing VSR countermeasures that will improve functional performance so that crewmembers will be able to meet early mission objectives during gravitational transitions. We envision that VSR stimulation will be available as a skin patch vestibular prosthesis during space flight that will further act synergistically along with the pre- and in-flight SA training to improve performance of functional tasks during gravitational transitions. This countermeasure approach can be implemented with minimal or no additional cost in terms of in-flight crew time, which is a significant advantage. The overlay of this countermeasure directly with any existing pre- or in-flight training modality to improve acquisition of skills will enhance the utility of the integrated, multi-disciplinary countermeasure capable for fulfilling multiple requirements making it a comprehensive and cost effective countermeasure approach. Earth-Based Applications: VSR stimulation prostheses have Earthbound application in rehabilitation of patients with balance disorders, strokes, spinal cord injury, peripheral neuropathy in the legs or a single hand that has been injured, and low vision, and for fall prevention training among seniors, as well as in both military and civilian occupations that involve shipping across oceans and flying in turbulence. This project will also enhance the efficacy of ground-based rehabilitation and training programs.

Task Progress:	The objective of two separate studies that were conducted was to document human visual performance during simulated low frequency vehicular motion. In the first study we examined changes in accuracy in normals when performing a seated visual target acquisition task in which the location of target was offset vertically during horizontal full-body rotation at an oscillating frequency of 0.8 Hz (peak velocity-160 deg/s). The main finding was that the accuracy of performance was degraded and response times varied with target location when acquiring off-plane targets at perturbing frequencies of 0.8 Hz in the horizontal plane. In the second study we examined 112 normals and 45 patients who had been diagnosed with either unilateral vestibular weaknesses or with post-acoustic neuroma resections. We determined their seated dynamic visual acuity (DVA) performance in stationary and during 5 cm vertical oscillations of 2 Hz. Scores were worse for both groups during the dynamic condition vs. the static condition. Further, scores for patients were significantly worse than normals in the dynamic condition. These results indicate that low frequency perturbations in the horizontal and vertical planes can cause decrements in visual performance and can exacerbate sensorimotor deficits after space flight. We studied the efficacy of VSR stimulation in three separate studies – In the first study VSR efficacy was evaluated during an otolith-canal conflict scenario in a variable radius centrifuge at low frequency oscillation (0.1 Hz) on physiological (eye movements) and perceptual (joystick) responses. Results show that VSR stimulation significantly reduced the phase difference between both the eye counter roll movements as well as the perceptual tracking responses with respect to the imposed tilt in gravito-inertial vector. These results indicate that VSR can improve otolith specific responses in intra-vestibular conflict scenarios like that experienced during space flight. In the second study VSR efficacy was evaluated on the cross-planar improvement in balance performance on an unstable surface while stimulating in the mediolateral (ML) axis only. Low imperceptible levels of white noise based binaural bipolar electrical stimulation of the vestibular system improved cross-planar balance performance. The amplitude of optimal stimulus for improving balance performance was predominantly in the range of 30-210 μA RMS. These results indicate that bipolar binaural stimulation may be sufficient to provide a comprehensive countermeasure approach for improving postural stability. In the third study VSR efficacy was evaluated during locomotion on an unstable surface. Low imperceptible levels of white noise based binaural bipolar electrical stimulation of the vestibular system improved locomotor performance consistent with the SR phenomenon in normal healthy subjects. These results indicate that VSR may be used to maximize locomotor performance during dynamic movements.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Articles in Peer-reviewed Journals	Peters BT, Mulavara AP, Cohen HS, Sangi-Haghpeykar H, Bloomberg JJ. "Dynamic visual acuity testing for screening patients with vestibular impairments." J Vestib Res. 2012 Jan 1;22(2):145-51. PubMed PMID: 23000614 , Jan-2012
Articles in Peer-reviewed Journals	Cohen HS, Mulavara AP, Peters BT, Sangi-Haghpeykar H, Bloomberg JJ. "Tests of walking balance for screening vestibular disorders." J Vestib Res. 2012 Jan 1;22(2):95-104. PubMed PMID: 23000609 , Jan-2012